

NATIONAL UNIVERSITY OF SINGAPORE

EXAMINATION FOR

(Semester II: 2020/2021)

EE5507 – ANALOG INTEGRATED CIRCUIT DESIGN

Apr / May 2021 - Time Allowed: 2.5 Hours

INSTRUCTIONS TO CANDIDATES:

1. This paper contains **FOUR** (4) questions and comprises **SEVEN** (7) printed pages.
2. Answer all **FOUR** (4) questions.
3. Each question carries 25 marks.
4. This is an OPEN BOOK examination.
5. Calculators can be used in the examination.
6. You may assume the following constant:

$$q=1.6021765 \times 10^{-19} \text{ C} \quad k=1.38064852 \times 10^{-23} \text{ m}^2 \text{ kg s}^{-2} \text{ K}^{-1}$$

Q.1

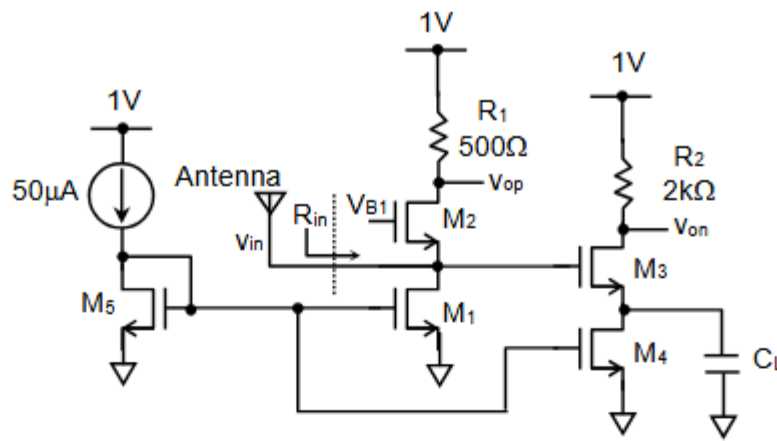


Fig. Q1

A single-ended to differential low noise amplifier (LNA) is shown in Fig. Q1. Assume all the NMOS body are connected to their source terminal respectively (No body effect). The supply voltage is 0.8V. The device parameters are given as follows:

- $L=130\text{nm}$: $V_{THN}=0.238\text{V}$, $\mu_n C_{ox}=386\mu\text{A/V}^2$, $\lambda_n=0.8\text{V}^{-1}$.
- $L=0.6\mu\text{m}$: $V_{THN}=0.238\text{V}$, $\mu_n C_{ox}=386\mu\text{A/V}^2$, $\lambda_n=0.02\text{V}^{-1}$.
- $(W/L)_{M5}=2.4\mu\text{m}/0.6\mu\text{m}$, $(W/L)_{M1}=24\mu\text{m}/0.6\mu\text{m}$, $(W/L)_{M4}=6\mu\text{m}/0.6\mu\text{m}$

- Determine the current for M_2 and M_3 . (2 marks)
- Determine the sizing of M_2 so that it will exhibit R_{in} of 50Ω . Assume L of 130nm . (5 marks)
- Determine the voltage gain (v_{op}/v_{in}). Determine the sizing of M_3 to exhibit similar gain, i.e. $|v_{on}/v_{in}|=|v_{op}/v_{in}|$. Assume L of 130nm . (5 marks)
- Determine minimum V_{B1} to ensure all transistors operate in saturation region. (4 marks)
- Suggest a simple modification to the circuit in Fig. Q1 such that the minimum V_{B1} can be reduced further. (4 marks)
- Determine the minimum supply voltage that the circuit can operate according to (d) (You may assume the signal is very small, and you do not have to consider the signal effect in your estimation). (5 marks)

Q.2

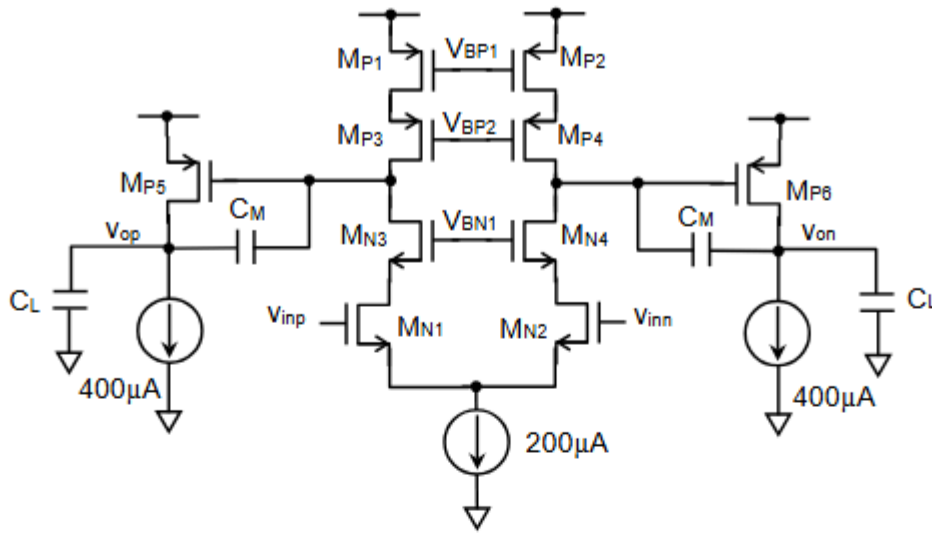


Fig. Q2a

Fig. Q2a shows a two-stage differential opamp. The common-mode feedback circuit is not shown and the supply voltage is 1.2V. You may assume the following:

- NMOS: $V_{THN}=0.238V$, $\mu_n C_{ox}=430\mu A/V^2$, $\lambda_n=0.8V^{-1}$.
 - PMOS: $V_{THP}=0.238V$, $\mu_p C_{ox}=110\mu A/V^2$, $\lambda_p=0.8V^{-1}$.
 - $(W/L)_{MN1}=(W/L)_{MN2}=(W/L)_{MN3}=(W/L)_{MN4}=1.3\mu m/130nm$.
 - $(W/L)_{MP1}=(W/L)_{MP2}=(W/L)_{MP3}=(W/L)_{MP4}=2.6\mu m/130nm$.
 - $M_{N1}, M_{N2}, M_{N3}, M_{N4}$: $g_{mn}=928\mu A/V$ and $r_{on}=9.8k\Omega$.
 - $M_{P1}, M_{P2}, M_{P3}, M_{P4}$: $g_{mp}=660\mu A/V$ and $r_{op}=5.8k\Omega$.
 - $C_L=10pF$, $C_M=2pF$
- a) Determine the sizing for M_{P5} and M_{P6} such that its transconductance is 3 times of M_{N1} and M_{N2} . Determine their output resistance as well. Assume minimum channel length of 130nm.
- (4 marks)**
- b) Estimate the overall gain, i.e. $(V_{op}-V_{on})/(V_{inp}-V_{inn})$.
- (4 marks)**
- c) Estimate the phase margin for feedback factor of 1, i.e. loop gain= $A(s)$ where $A(s)$ is the opamp open loop transfer function.
- (6 marks)**
- d) Estimate new C_M such that the phase margin is 50° .
- (6 marks)**

- e) One of the key issue of the above opamp circuit is limited input common mode range. Consider the inverting amplifier shown in Fig. Q2b, discuss the impact due to the above issue.

(5 marks)

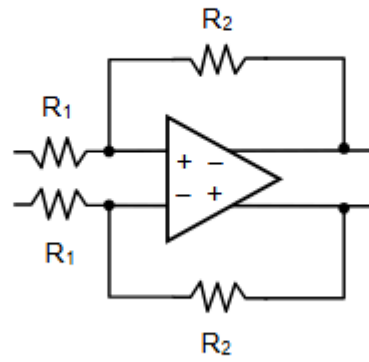


Fig. Q2b

Q3. Data Converter + Filter

- a) For the system shown below (Fig. Q3a), what is the minimum sampling frequency f_s that guarantees > 60 dB attenuation of aliasing components in the discrete time signal $y(n)$?

(4 marks)

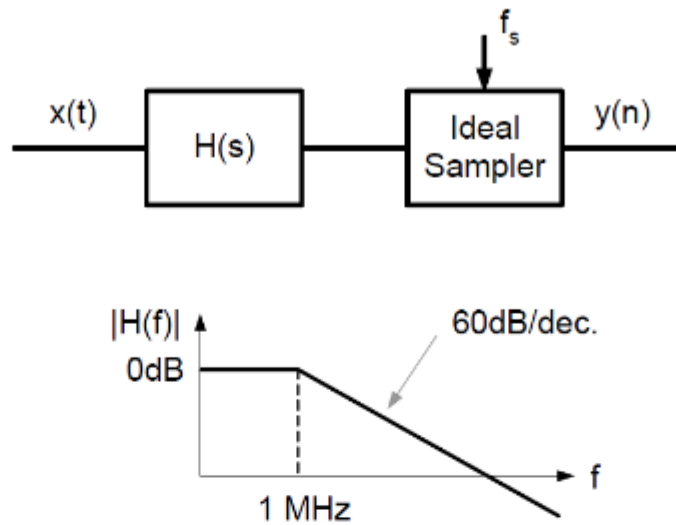


Fig. Q3a

- b) For a first order $\Delta\Sigma$ modulator with ideal 4-bit quantizer and 4-bit DAC, what is the oversampling ratio (M) required to obtain SQNR equivalent to a 14-bit ADC? (6 marks)
- c) The $\Delta\Sigma$ modulator shown in Fig. Q3b employs a B -bit ADC for increased resolution, but only 1-bit DAC is used to avoid distortion due to DAC nonlinearity. The remaining $B-1$ bits serve as an estimate of the quantization error. The advantage here is that the multi-bit quantizer has less quantization noise compared to single-bit quantizer and less oversampling is needed to achieve desired SQNR.
- i) Draw a linearized schematic that models the quantization error $E_1(z)$ and truncation error $E_2(z)$ as additive white noise sources. E_1 is quantization error of ADC. E_2 is truncation error due to non-MSB bits.
Hint: if output of ADC is $W(z)$, then $W(z) = \text{MSB} + E_2(z)$ (4 marks)
- ii) Derive the noise transfer function and signal transfer function of this modulator. What is the optimal value for G ? What happens to $E_2(z)$ at optimal value of G ? (5 marks)
- iii) What is the optimal SQNR of this modulator as a function of B and the oversampling ratio M ? Assume optimal choice of G and an ideal brickwall decimation filter. (3 marks)

- <https://claude.ai/share/c178499b-d242-405d-b31c-6cdca9a71a39>

- chat says the linearised schematic means the truncation is non linear, so linearise it by using truncation error to model it

- iv) Comment on the potential limitations of this architecture.

Hint: Consider how the analog and digital blocks of the modulator achieves the optimal SQNR.

(3 marks)

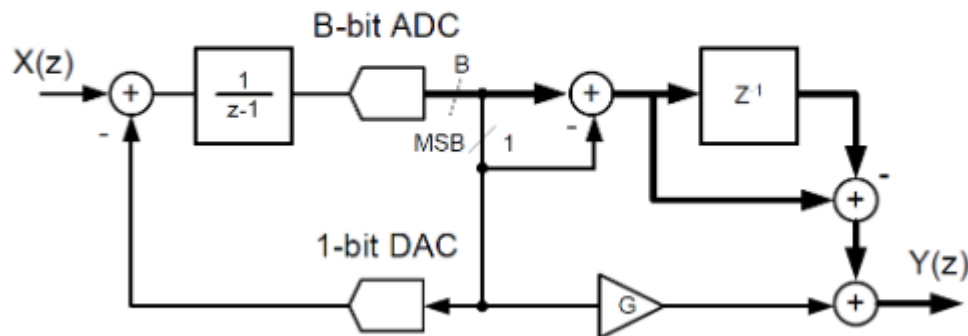


Fig. Q3b

Q4. Noise

- a) A white noise signal with root spectral density of $10\text{nV}/\sqrt{\text{Hz}}$ is applied to the input of a first-order lowpass RC filter with $R=1\text{k}\Omega$ and $C=0.159\mu\text{F}$. Including the thermal noise of the resistor, what is the total rms noise value at the output of the filter? Assume $T=300\text{K}$.

(5 marks)

- b) The output noise of a circuit is measured to be -20dBm around 100kHz when a resolution bandwidth of 40 Hz is used. Find the root spectral density in $\text{V}/\sqrt{\text{Hz}}$ around 100kHz . (**Hint:** $R=50\Omega$)

(5 marks)

- c) Find the output noise (in dBm and V_{rms} reference to 50Ω resistor) when 2 noise sources of -20dBm and -23dBm are added together.

(5 marks)

- d) Fig. Q4 shows a first-order RC filter. Estimate the expected signal-to-noise ratio (SNR) in dB for a 10mV_{rms} input signal with bandwidth up to 10kHz . State any assumptions you use.

You may assume opamp input noise $V_n(f) = 20\text{nV}/\sqrt{\text{Hz}}$, $I_{n+}(f) = I_{n-}(f) = 20\text{pA}/\sqrt{\text{Hz}}$, and unity gain frequency = 10MHz .

Hint: you need to consider all circuit noise sources to calculate noise at V_{out}

(10 marks)

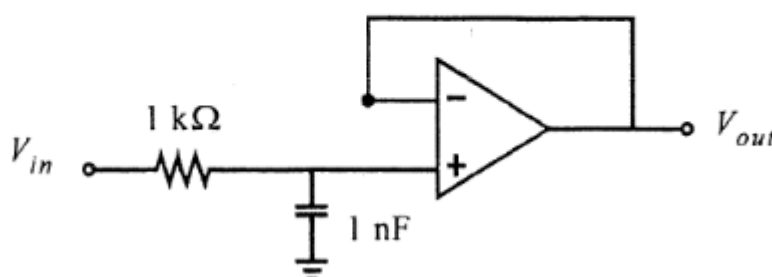


Fig. Q4